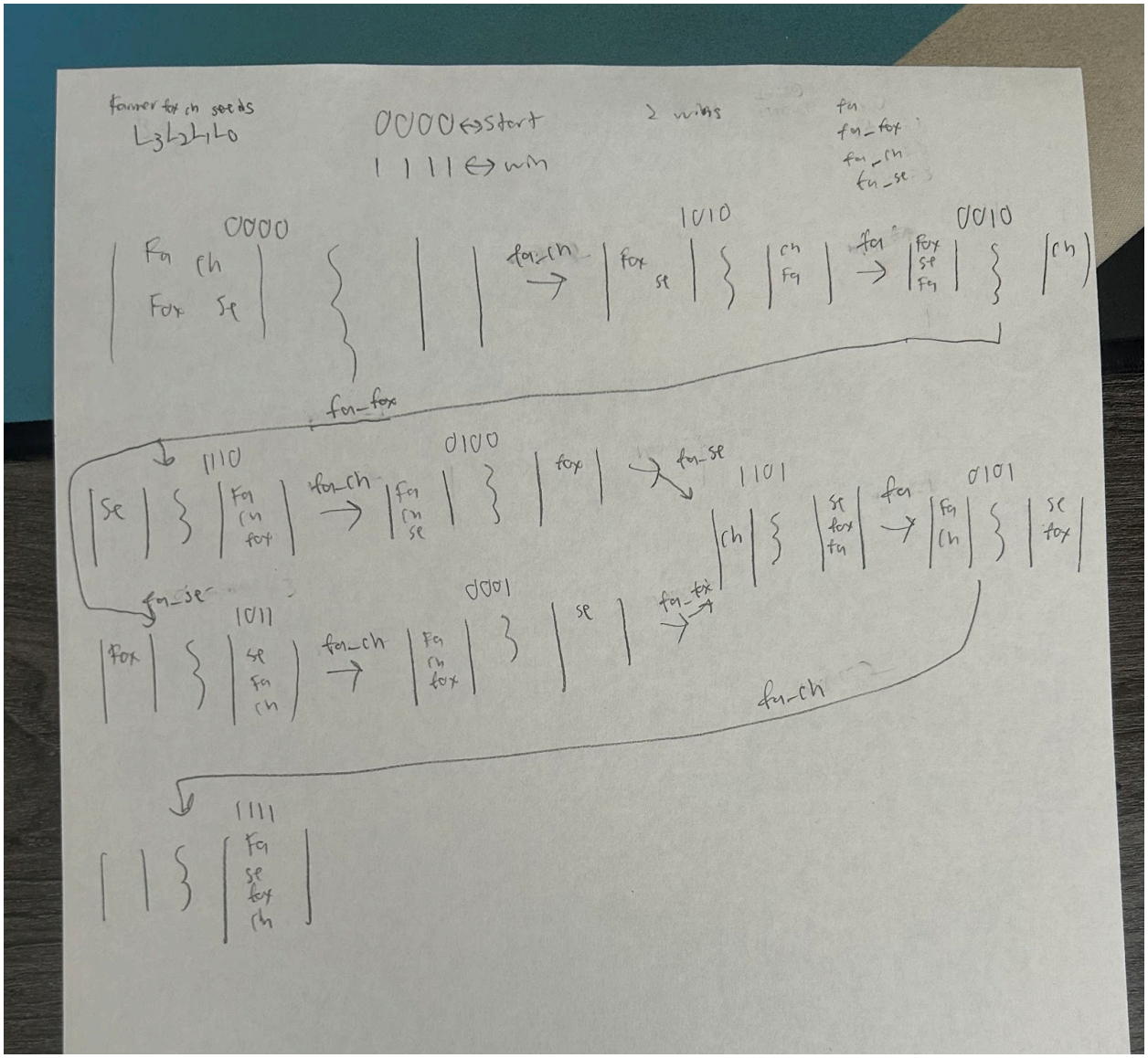
Dylan Thornburg ECEN 127 Lab 5 Prelab

PRE-LAB

(i) Given the description above, particularly the discussion of the inputs and outputs, define the interface for a Verilog module. This will require you to give explicit names to signals. If you define any of the signals as buses, make sure the width is clear.

```
module game(
        input clk, rst,
        input [2:0] move,
        output reg [3:0] state,
        output reg win, lose, impossible
);
```

(ii) Create a state diagram that maps out the state transitions. Remember that “impossible” moves should keep the machine in the same state. And once the machine reaches the winning state or any of the losing states, it should stay where it is until the reset signal is asserted. Make sure to be clear which outputs are Moore-style and which are Mealy-style. To keep the diagram reasonably clean, you don’t need to show reset. Note that having the state machine coded in Verilog is not a requirement for pre-lab, but it is highly recommended that you have your code entered and ready to test before arriving at lab.



State diagram only shows win paths as any loss simply ends game and keeps you perpetually in one of the 4 lose states.

(iii) Outline a handful of important testcases that you will run against your design to make sure it's working properly. You will need to write the complete testbench.

Test both win paths.
Make sure impossible keeps the state the same until new input.
Make sure win and lose keep the state the same until rst is asserted regardless of input changes.